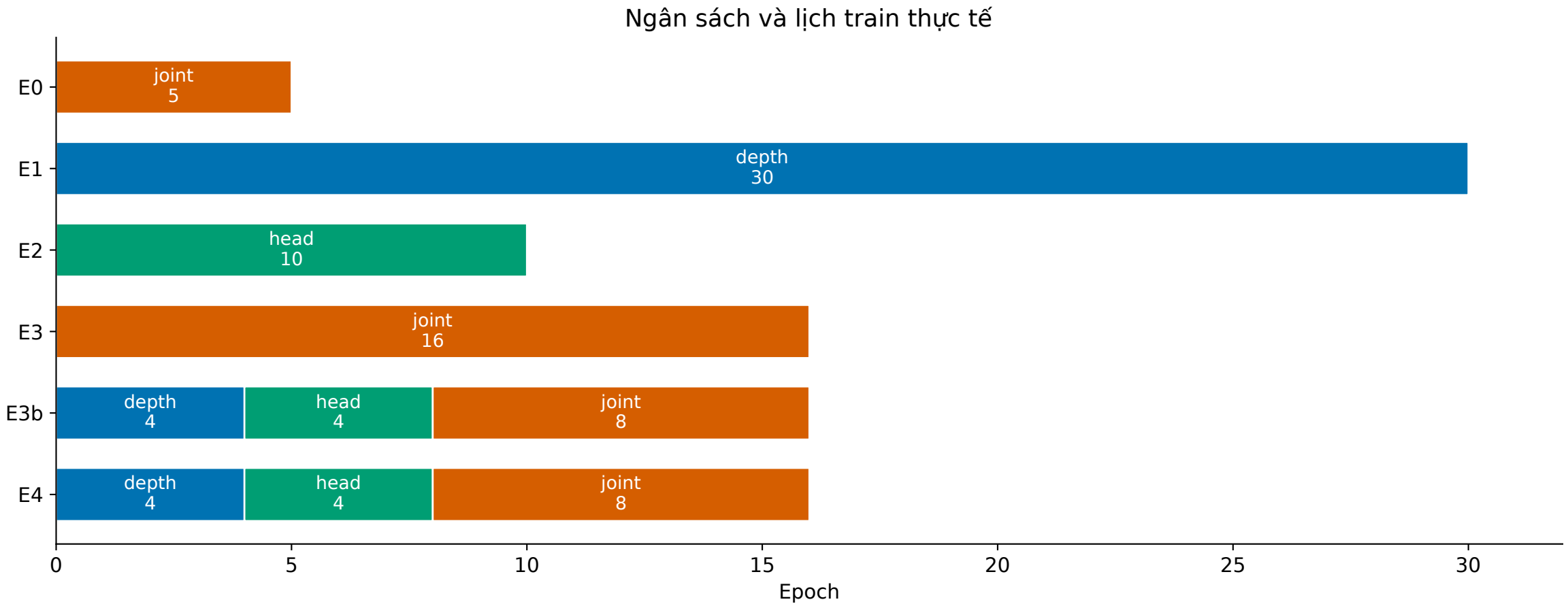

CASIA development protocol · seed 42 · video mean of 20 frames · 270 attack / 90 live test videos. E0 joint means binary BCE training;
E1 depth-only. E0/E1/E2 budgets differ from the 16-epoch E3/E3b/E4 ablation.